

Kubernetes Lab: NodePort Service

Trainer: Sandeep Kaushik

Learning Objective

Understand how a NodePort Service exposes an application running inside Kubernetes to external users.

Learning Outcomes

By the end of this lab, participants will be able to: - Deploy an application using Kubernetes Deployment - Expose the application using NodePort Service - Access the application using Node IP and NodePort - Understand how external traffic reaches Pods

Lab Environment Requirement

- Kubernetes Cluster
 - kubectl configured
 - Access to node IP
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Step 1: Create Deployment

Create a file named **deployment.yaml**

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-nodeport-demo
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx-demo
  template:
    metadata:
      labels:
        app: nginx-demo
    spec:
      containers:
        - name: nginx
```

```
image: nginx
ports:
- containerPort: 80
```

Apply the deployment

```
kubectl apply -f deployment.yaml
```

Verify pods

```
kubectl get pods -o wide
```

Step 2: Create NodePort Service

Create a file named **service.yaml**

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-nodeport-service
spec:
  type: NodePort
  selector:
    app: nginx-demo
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30008
```

Apply the service

```
kubectl apply -f service.yaml
```

Verify the service

```
kubectl get svc
```

You should see something similar to:

NAME	TYPE	CLUSTER-IP	PORT(S)
nginx-nodeport-service	NodePort	10.x.x.x	80:30008/TCP

Step 3: Get Node IP

```
kubectl get nodes -o wide
```

Example:

```
192.168.1.2
```

Step 4: Access the Application

Open a browser or run:

```
curl http://192.168.1.2:30008
```

You should see the NGINX welcome page.

This confirms that the NodePort Service is exposing the application externally.

How NodePort Works

When a request reaches the node on port **30008**, Kubernetes forwards that request to the Service. The Service then distributes the request to one of the running Pods.

Traffic Flow:

User → NodeIP:NodePort → Service → Pod

Default NodePort Range

By default Kubernetes assigns NodePorts in the following range:

30000 - 32767

This range is used to avoid conflicts with common system ports like HTTP (80), HTTPS (443), and SSH (22).

Key Takeaway

NodePort allows external users to access applications running inside the Kubernetes cluster by exposing a port on every node in the cluster.